

Evaluating The Evaluator

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Abstract

Automated evaluation of logical scientific content is a significant and unresolved challenge. In this paper, we apply multiple existing evaluation methods on a specific dataset that consists of questions and answers from the Data Structures Course in TAU, and assess the quality of their evaluation. We examine methods that utilize common NLI models and methods based on LLMs (such as GPT) as the evaluator. We conclude that NLI models with their current training are not sufficient evaluators and that LLM-based evaluation currently presents better performances. From the results, we form the conjecture that a sufficient evaluation method of the dataset could be feasible with further research and training.

1 Introduction

Establishing an unbiased, automated evaluation that correlates well with human judgment for long, heavily logical scientific content is an emerging field of research (Shaham et al., 2022). In this paper, we focus on evaluating solutions for questions from the Data Structures course in TAU. These questions are open-ended questions, meaning their solutions contain elaborate explanations and a number of logical steps, thus making it a challenge to evaluate their correctness. This gives us hope that the insights we gain from examining this dataset will apply to similar scientific and logical data.

The solutions we evaluate in this paper were generated by GPT-3.5 Turbo. Evaluating solutions generated by an NLG (Natural Language Generation) is practical for two reasons; First of all, using NLG is an efficient way to generate enough correct and incorrect solutions to hold a reliable experiment and to gain stable statistics; Second of all, assessing Large Language Models' (LLMs') correctness and accuracy is a major research challenge, and this paper could be used to create a benchmark

that ranks LLMs on their ability to generate longer scientific content.

Finding a Good Evaluation Methods A good evaluation method should be automatic and accurate. Such a method could be used for a benchmark for LLMs, and thus help advance them. It could also be helpful in many educational scenarios: professors grading exams, students who want to self-evaluate themselves, and many more.

In our experiment, we explore various leading automatic evaluation methods on the solutions to the data structure questions dataset and we try to identify which method would be most accurate on this dataset. Even though this dataset contains questions gathered from a specific course, we would like to believe that the results we achieve might be applicable to evaluate other data in fields that have similar characteristics to these questions and answers, such as long output, question answering, logic inference and coherence and algorithm building.

2 Related Work

We examine two categories of evaluators, in each we tried several specific methods.

NLI Models The first category utilizes an NLI (Natural Language Inference) model to compare pieces of text. NLI models usually expect to receive a premise and a hypothesis, and they output entailment / contradiction / neutral based on their perceived relation between the two pieces of text.

This technique has a natural requirement that suits our circumstances. The dataset we are testing includes an official solution (written by the teacher and therefore correct), which makes it natural to compare the NLG's output answer to the official solution in order to evaluate the generated answer. If the generated answer is correct, then we expect the official solution and the generated answer to entail

each other, and vice versa, if both solutions entail each other, we hypothesize the generated answer is correct. NLI models were created to determine the logical relationship between two pieces of text, making them a useful tool to conduct the logical entailment test mentioned above.

We examined a DeBERTa-based NLI (He et al., 2021a) (He et al., 2021b), a Bart-based NLI (Lewis et al., 2019), and a T5-based NLI model (Honovich et al., 2022).

LLM-Based Methods The second category is sometimes referred to as “LLM as a judge”. In general, this method is based on using an open LLM model such as GPT as the evaluator. To use this method, one must provide the evaluator with instructions and evaluation criteria, and the data to evaluate. This method was explored in the paper (Liu et al., 2023). In their research, they compared several evaluation methods on two NLG tasks, summarization and dialogue response generation. They concluded that this method outperforms all the other methods they examined.

This method has some major inherent disadvantages; For one, when using an open LLM model such as GPT, results may be inconsistent and some parts of the logic of the evaluator remain unknown and possibly unstable. In addition, as specified in the previously mentioned work, LLMs as evaluators have a bias towards LLM generated text, which might affect the evaluators accuracy. This bias could influence our experiment, as the solutions we aim to evaluate were generated by an LLM.

An additional evaluation method we read about is called SelfCheckGPT (Manakul et al., 2023). This paper used the consistency of multiple outputs generated by using the same prompt multiple times, in order to evaluate the LLM’s output. Their work shows that if a generated response is consistent, then it is more likely that it is correct. This method is exceptionally useful when the output is short and determined, but the desired output is unknown or is not included in the dataset. In the paper they show that this method works well for fact checking, showing that response consistency may imply accuracy. Although in our experiment we did not examine this method, it is a significant idea that raises the importance of consistency in LLMs’ responses.

3 Methods

3.1 Dataset

The dataset we use is a data frame that represents questions from the final exams of the Data Structures course in TAU, where each row is a separate question. The data frame consists of the following columns:

exam_year, *semester*, *moed*, *url*, *question_number* - meta data on the question, these fields are sufficient to uniquely identify each question.

question_index - is the question’s index in the original data frame received from the staff.

question_type - B/C for short answers and A for long answers.

question_translation_latex - the question itself, written in English and latex format.

question_in_hebrew - the original Hebrew question.

answer_translation_latex - the official solution written in English and latex format.

answer_in_hebrew - the original Hebrew official solution.

3.2 Experimental Setup

The first step was to pick the questions that will be included in our dataset. All the questions we picked are from types A/B/C, they have clear official solutions that are worth comparing to in order to evaluate, and they were already tested in previous work of other students.

The second step was to prompt GPT-3.5 Turbo, to solve the questions. The answer to each question appears in the column *chatgpt_answer* in the data frame.

The prompt we used -

Answer the following question. Write the solution in Latex format:

{question_translation_latex}

The third step was to manually evaluate the output generated by GPT-3.5 Turbo, meaning, we read each of the generated solutions and determined ourselves if they are correct or not. This stage is necessary for evaluating the evaluation methods. After completing this step, we found that GPT-3.5 Turbo correctly answered 32% of the questions, and answered incorrectly on 68% of the questions. Having both correct and incorrect answers will enable the evaluators to tackle both tasks of identifying correct and incorrect answers. Because we

want to evaluate the performance of the evaluators on both tasks, some of the results are presented separately for each task.

The fourth step was to experiment with multiple evaluation methods, all based on comparing the official solution and the GPT generated solution in order to decide whether the generated solution is correct. We then compared each method’s labeling of "correct" and "incorrect" to the ground truth we gained by the manual evaluation. For each question, we say that an evaluator is correct if one of the following conditions is met -

- The generated solution was indeed correct (by the manual evaluation) and the evaluator identified it as correct.
- The generated solution was incorrect (by the manual evaluation) and the evaluator identified it as incorrect.

3.3 Evaluators Examined

Since NLI models are self contained and consistent, we started by examining the following SOTA models, which were trained on the datasets described in the next section.

- cross-encoder/nli-deberta-v3-base - This model is based on microsoft/deberta-v3-base (He et al., 2021a) (He et al., 2021b) and trained on the SNLI and MultiNLI datasets.
- facebook/bart-large-mnli - This model is based on facebook/bart-large (Lewis et al., 2019) and trained on the MultiNLI dataset.
- google/t5_xxl_true_nli_mixture - This model is based on T5-XXL. It is trained similarly to the NLI model described in the TRUE paper (Honovich et al., 2022), but using the following datasets instead of ANLI: SNLI, MultiNLI, Fever, Scitail, PAWS and VitaminC.

Datasets for Training NLI Models

- SNLI (Bowman et al., 2015) consists of 570K labeled sentence pairs, written by humans based on image captioning.
- MultiNLI (Williams et al., 2018) consists of 433K labeled examples from ten distinct genres of written and spoken English.

- Fever (Thorne et al., 2018) is a dataset for verification against textual sources, consisting of 185K claims generated by altering sentences extracted from Wikipedia.
- Scitail (Khot et al., 2018) is a dataset and model for textual entailment, derived from treating multiple-choice question-answering as an entailment problem.
- PAWS (Zhang et al., 2019) consists of 108K pairs of sentences with high lexical overlap.
- VitaminC (Schuster et al., 2021) consists of 400K claim-evidence pairs that are nearly identical in language and content, with the exception that one supports a given claim while the other does not.

NLI-Based Evaluation

With all NLI models, we examined both directions of inference (official solution $\implies$ generated answer, generated answer $\implies$ official solution). With Bart and DeBERTa we completed two different experiments - the first was comparing just the solutions, and the second was comparing a longer text that included the question itself and the solution, meaning the premise and hypothesis were as following:

```
premise = "question: "
+ row['question_translation_latex']
+ "solution: "
+ row['answer_translation_latex']
hypothesis = "question: "
+ row['question_translation_latex']
+ "solution: "
+ row['chatgpt_answer']
```

The purpose of adding the question to the compared text was to give the evaluator more context. The added context seemed significant since deciding the relation between two solutions without knowing the question is not a trivial task, even for a human.

A remark on the T5-based evaluator - unlike the other models, this model has only two output types - “entailment” and “no entailment”. This affects the results, as this model has no "neutral" option.

LLM-Based Evaluation

Since the results of the NLI-based methods (further discussed in the “Results” section) were not satisfying, we examined two more methods, both

are based on an LLM as the evaluator, as was mentioned in the “Related Work” section, using comparison to the official solution as the main evaluation criteria. In the following methods, since the LLM we used (GPT-4o) is not completely consistent, each experiment was preformed 5 times, see Table 1 for further explanation.

GPT-4o Grader This method is loosely based on the G-EVAL method (Liu et al., 2023). Its main concept is that the LLM receives both the official solution and the GPT generated answer, and is prompted to compare them and output a grade for the generated answer. We examined this method using 3 different grading scales: 1-4, 1-5, 1-10.

GPT settings -

model="gpt-4o",
temperature = 0

Prompt used -

You are a computer science professor at the university.

You will be given two solutions to a question from a test. The first is the teacher solution and it is correct for sure, The second is a student solution that needs to be graded.

Please check the student solution. Give it a grade between 1 and 4, where 1 is for a very incorrect answer and 4 is for a great answer.

You are only allowed to answer 1 or 2 or 3 or 4, write just the grade without any additional text.

Teacher solution:

{answer_translation_latex}

student solution:

{chatgpt_answer}

GPT-4o as NLI This method is based on the concept of using LLMs for different NLP tasks (Sun et al., 2023). Its main concept is to simulate an NLI using GPT-4o, meaning prompting the LLM to act as an NLI. Our purpose with this method was to speculate whether an LLM can perform the task of inference on our type of data.

GPT settings -

model="gpt-4o",
temperature = 0

Prompt used (similar for contradict / neutral) -

*Does the premise entail the hypothesis?
Please answer just yes or no without any additional text.*

Premise:

{answer_translation_latex}

Hypothesis:

{chatgpt_answer}

Question Types Tested

An important remark on the type of questions examined - questions of type A (long answers) were not included in the experiments with the NLI models, only of types B and C (short answers) were used. This is because we assume that questions of type B / C are easier to evaluate, and since the evaluation on them was not satisfying, there was no use in examining them on questions of type A. LLM-based evaluation methods were examined on questions of types A, B and C.

4 Results

4.1 Measurements Taken

Accuracy out of all Out of all questions, how many were labeled correctly?

Decided out of all Out of all questions, how many were decided? This measurement does not directly reflect the quality of the model’s assessment but it does affect other measurements.

Accuracy out of decided Out of all the questions the evaluator decided on a label to their GPT generated answer (not neutral), how many were labeled correctly?

Precision of the “correct” labeling Out of all the questions that their GPT generated answers were labeled as “correct”, how many were actually correct?

Precision of the “incorrect” labeling Out of all the questions that their GPT generated answers were labeled as “incorrect”, how many were actually incorrect?

Recall of actually correct answers Out of all the questions that their GPT generated answers are actually correct, how many were labeled by the evaluator as correct?

Recall of actually incorrect answers Out of all the questions that their GPT generated answers are actually incorrect, how many were labeled by the evaluator as incorrect?

4.2 Results Analysis

Table 1 shows the transition from raw output to labeling.

Table 2 shows all methods’ results on questions of type B / C. As mentioned before, NLI models were not examined on questions of type A because of their low performance on questions of type B / C.

Table 3 shows GPT-4o Grader and GPT-4o as NLI results on questions of type A / B / C

NLI Results Analysis In general, recall rates are very unsatisfying, meaning, we assume that on new data, if one knows what the correct labeling should be, the probability to actually get that label is very low. This result is related to the fact that all models could not decide (return "neutral" label) on many questions. In addition, when the models do decide on a label, their accuracy rates are not sufficient and are close to a coin flip. Specifically on T5, the recall rates for truly incorrect answers seem high, but that is because the model has a tendency to label as “incorrect”. This can be seen in the unsatisfying result for the recall of actually correct answers.

When adding the question to the text used for comparison, we can observe that it did not improve any measurements, except for the recall of actually correct answers. This result is not surprising, since we saw an increment in “correct” labeling of the GPT generated answers by the evaluators. We suspect that this is related to the fact that when the question is a part of both the texts that are compared, they seem similar as they share a common paragraph.

LLM-Based Methods Results Analysis In general we can see an improvement in all measurements compared to the NLI evaluators. Specifically the precision of “incorrect” labeling improved, meaning that if the label returned by the evaluator was the “incorrect” label, it can be somewhat trusted.

Comparing the 1-4 and 1-5 graders, we noticed that the first is more “decisive”, and the latter is a bit more accurate. A possible explanation for this is that on a scale of 1-4, there is no “middle” or natural “neutral” choice. On the contrary, a scale

of 1-5 has a natural “neutral”, making the decided questions more pronounced.

5 Discussion

Our first conclusion from our experiment is that existing general NLI models, with their current training data, are not good enough for comparing solutions and checking correctness for the types of questions and solutions we examined them on. Using an LLM as an evaluator that compares solutions, with both methods we tested (as a grader and as an NLI), produced better results. Even though the results were better, and might be somewhat trustworthy in some cases, such as in precision of incorrect labeling, there are still gaps that make it difficult to trust LLMs as automatic evaluators. Nonetheless, the improvements we saw using an LLM as an evaluator, show feasibility of using LLMs as NLI evaluators.

As for the measurement of precision of incorrect labeling using LLM-based evaluation methods, it seems consistently better than other measurements. A possible explanation for this measurement being distinct from the others is that even for a human, deciding that a solution is wrong is easier than deciding it is correct. The first task only requires finding a single mistake, while the second task requires verifying there are no mistakes at all.

6 Future Work

Statistics on questions of type A were not presented separately as we only have a small sample of them. It seems that evaluation of questions of type A, which is a harder task for humans than evaluation of questions of type B / C, is also harder for the LLMs, as adding them to the statistics lowered their perceived performance. Future work could test this hypothesis.

Because GPT-4o outperformed the general NLI models we tried, we conclude that LLMs in general can perform this task. Therefore we assume it could be possible to train and fine-tune NLI models to perform better on this type of data comparison, and achieve better results using them, while still gaining the closed NLI models’ advantages that were mentioned before.

In this work, we did not try many different prompts for the evaluation. It is possible that other prompts could lead to even better results.

Another closing remark is that this work aspired to find a binary evaluation method. There are use

Model	Label “incorrect”	Label “neutral”	Label “correct”
DeBERTa	[contradict, contradict] [contradict, neutral] [neutral, contradict]	[neutral, neutral]	[entail, entail] [entail, neutral] [neutral, entail]
BART	[contradict, contradict] [contradict, neutral] [neutral, contradict]	[neutral, neutral]	[entail, entail] [entail, neutral] [neutral, entail]
T5	[0, 0]	Any other output	[1, 0] [0, 1] [1, 1]
GPT-4o			
- Grader 1-4	1-2	2-3	3-4
- Grader 1-5	1-2.5	2.5-3.5	3.5-4.5
- Grader 1-10	1-4.5	4.5-6.5	6.5-10
GPT-4o as NLI	returned “contradict” and did not return “entail”	returned “neutral” and did not return “contradict” or “entail” or returned “contradict” and “entail”	returned “entail” and did not return “contradict”

Table 1: Transition from raw output to labeling. Labeling in GPT-4o Grader: We repeated the experiment 5 times and the final grade is the average of the results. Labeling in GPT-4o as NLI: In this experiment GPT was prompted 3 yes or no questions - for entail, contradict and neutral. We repeated the questions 5 times and decided by the majority.

Model	Accuracy out of all (%)	Decided out of all (%)	Accuracy out of decided (%)	Precision of the "correct" labeling (%)	Precision of the "incorrect" labeling (%)	Recall of actually correct answers (%)	Recall of actually incorrect answers (%)
DeBERTa	22.9	40	57.1	46.7	69.2	25.9	20.1
- with question	25.7	65.7	41.3	39	50	59.2	4.6
Bart	25.7	67.1	38.2	30.7	71.4	29.6	23.2
- with question	26.8	74.6	35.8	35.5	47	39.3	18.6
T5	57.5	97.5	59	38.4	62.1	17.2	80.3
GPT-4o							
- Grader 1-4	75.6	92.7	81.6	76.9	84	66.7	80.8
- Grader 1-5	59.8	69.5	86	83.3	87.1	50	65.4
- Grader 1-10	70.7	89	79.4	71.9	85.4	76.7	67.3
GPT4-o as NLI	48.8	62.2	78.4	76.9	78.9	33.3	57.7

Table 2: All methods’ results on questions of type B / C

Model	Accuracy out of all (%)	Decided out of all (%)	Accuracy out of decided (%)	Precision of the "correct" labeling (%)	Precision of the "incorrect" labeling (%)	Recall of actually correct answers (%)	Recall of actually incorrect answers (%)
GPT-4o							
- Grader 1-4	70.5	89.5	78.7	65.6	85.5	63.6	73.6
- Grader 1-5	53.3	64.8	82.4	69.6	88.9	48.4	55.6
- Grader 1-10	60.9	82.8	73.5	59.5	86.7	75.8	54.1
GPT-4o as NLI	48.6	61	79.7	66.7	83.7	30.3	56.9

Table 3: GPT-4o Grader and GPT-4o as NLI results on questions of type A / B / C

cases where a more fine evaluation is required, such as grading a student’s exam. Future work could focus on trying to find a more fine and precise type of evaluation.

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